

Twisted States in Higgs Boson Decay into Fermion Antifermion Pair

Borodin Nikita

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1 Introduction

Problem Statement

Consider the decay of the Higgs boson into a pair of fermions $f\bar{f}$. The quantum state of the fermions is given by:

$$|f\bar{f}\rangle = \int \frac{d\vec{p}_1}{(2\pi)^3\sqrt{2E_1}} \frac{d\vec{p}_2}{(2\pi)^3\sqrt{2E_2}} \psi(\vec{p}_1, s_1; \vec{p}_2, s_2) |f(\vec{p}_1, s_1), \bar{f}(\vec{p}_2, s_2)\rangle,$$

where ψ is the two-particle wave function of fermions in momentum representation, s_i are the spin 4-vectors. In the rest frame of the fermion $s = (0, \boldsymbol{\xi})$.

1. Find ψ using the matrix element:

$$\langle f(\vec{p}_1, s_1), \bar{f}(\vec{p}_2, s_2) | S - 1 | \text{Higgs boson} \rangle.$$

2. Now one can calculate the wave function in coordinate representation:

$$\psi(\vec{x}_1, \vec{x}_2) = \int \frac{d\vec{p}_1}{(2\pi)^3} \frac{d\vec{p}_2}{(2\pi)^3} \tilde{\psi}(\vec{p}_1, s_1; \vec{p}_2, s_2) e^{+i(\vec{p}_1\vec{x}_1 + \vec{p}_2\vec{x}_2)}.$$

Show that $\psi(\vec{x}_1, \vec{x}_2)$ depends on the coordinate difference $\vec{r} = \vec{x}_1 - \vec{x}_2$ and calculate the wave function $\psi(\vec{r})$.

3. Expand the wave function in a series of spherical harmonics and investigate their behavior as a function of $\boldsymbol{\xi}_1 \cdot \hat{\vec{r}}, \boldsymbol{\xi}_2 \cdot \hat{\vec{r}}, \boldsymbol{\xi}_1 \cdot \boldsymbol{\xi}_2$. Give an interpretation.

2 Find wavefunction

Using the Fourier transform

$$\begin{aligned} \psi(\vec{r}_1, \vec{r}_2) &= \int \frac{d^3p_1 d^3p_2}{(2\pi)^6 \sqrt{2E_1 2E_2}} \tilde{\psi}(\vec{p}_1, \vec{p}_2) e^{i(\dots)} \\ &= \int \frac{d^2p_1 d^2p_2}{(2\pi)^3} \langle f\bar{f} | S - 1 \rangle e^{i(\dots)} \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} \frac{m_f}{v} \bar{u}(\vec{p}_1, \vec{s}_1) u(-\vec{p}_2, \vec{s}_2) e^{i(\dots)} \\ S | i \rangle &= | f \rangle = \dots + \int \frac{d^3p_1 d^3p_2}{(2\pi)^6} \frac{\tilde{\psi}(\vec{p}_1, \vec{p}_2)}{\sqrt{2E_1 2E_2}} | \vec{p}_1, \vec{s}_1; \vec{p}_2, \vec{s}_2 \rangle + \dots \end{aligned}$$

$$\langle \vec{p}_1 \vec{p}_2 | \vec{p}_1 \vec{p}_2 \rangle = (2\pi)^3 \sqrt{2E_1 2E_2} [\delta^3(\vec{p}_1 - \vec{p}_2) \delta'(\vec{p}_2 - \vec{p}_1') - \delta^3(\vec{p}_1 - \vec{p}_2) \delta'(\vec{p}_2 - \vec{p}_1)]$$

Momentum

$$\begin{aligned} |p\rangle &\equiv \sqrt{2E_p} a_p^\dagger |0\rangle \\ \langle \vec{q} | \vec{p} \rangle &= \langle \vec{q} | a_q \sqrt{2E_q} \sqrt{2E_p} a_p^\dagger |0\rangle = 2E_p (2\pi)^3 \delta^3(\vec{q} - \vec{p}) \\ a_q a_p^\dagger &= -a_q^\dagger a_p + \underbrace{\{a_q, a_p^\dagger\}}_{(2\pi^3)\delta^3(\vec{q}-\vec{p})} \end{aligned}$$

We'll get the following norm

$$\langle \vec{q}, s | \vec{p}, r \rangle = 2E_p (2\pi)^3 \delta^3(\vec{q} - \vec{p}) \delta_{sr}$$

Substitute in our decay

$$\begin{aligned} \langle q_1, r_1; q_2, r_2 | p_1 s_1; p_2 s_2 \rangle &= \sqrt{2E_{p_1} 2E_{p_2} 2E_{q_1} 2E_{q_2}} \langle 0 | a_{q_1 r_1} a_{q_2 r_2} a_{p_1 s_1}^\dagger a_{p_2 s_2}^\dagger | 0 \rangle \\ &= \sqrt{2E_{p_1} 2E_{p_2} 2E_{q_1} 2E_{q_2}} \langle 0 | a_{q_1 r_1} (-a_{q_2 r_2}^\dagger a_{p_1 s_1} + \{a_{q_2 r_2}, a_{p_1 s_1}^\dagger\}) a_{p_2 s_2}^\dagger | 0 \rangle \\ &= \{a_{q_2 r_2}, a_{p_1 s_1}^\dagger\} \langle 0 | a_{q_1 r_1} a_{p_2 s_2}^\dagger | 0 \rangle - \langle 0 | a_{q_1 r_1} a_{q_2 r_2}^\dagger a_{p_1 s_1} a_{p_2 s_2}^\dagger | 0 \rangle \\ &= \{a_{q_2 r_2}, a_{p_1 s_1}^\dagger\} \{a_{q_1 r_1}, a_{p_2 s_2}^\dagger\} - \{a_{q_1 r_1}, a_{q_2 r_2}^\dagger\} \{a_{p_1 s_1}, a_{p_2 s_2}^\dagger\} \end{aligned}$$

Finally will get

$$\begin{aligned} \langle \vec{q}_1 r_1; \vec{q}_2 r_2 | S | H \rangle &= \int \frac{d^3 p_1 d^3 p_2}{(2\pi)^6 \sqrt{2E_{p_1} 2E_{p_2}}} \tilde{\psi}(\vec{p}_1, \vec{p}_2) \langle q_1 r_1; q_2 r_2 | p_1 s_1; p_2 s_2 \rangle \\ &= \int \frac{d^3 p_1 d^3 p_2}{(2\pi)^6 \sqrt{2E_{p_1} 2E_{p_2}}} \tilde{\psi}(\vec{p}_1, \vec{p}_2) [(2\pi)^6 \delta^3(q_2 - p_1) \delta_{r_2 s_1} \delta^3(q_1 - p_1) \delta_{r_2 s_1} - (2\pi)^6 \delta^3(q_2 - p_2)] \\ &= \frac{\tilde{\psi}(q_2, q_1)}{\sqrt{2E_{q_1} 2E_{q_2}}} \delta_{r_2 s_1} \delta_{r_1 s_2} - \frac{\tilde{\psi}(q_1, q_2)}{\sqrt{2E_{q_1} 2E_{q_2}}} \delta_{r_1 s_1} \delta_{r_2 s_2} \end{aligned}$$

??

$$\frac{\tilde{\psi}(q_2 r_2; q_1 r_1) - \tilde{\psi}(q_1 r_1; q_2 r_2)}{\sqrt{2E_{q_1} 2E_{q_2}}}$$

$$\tilde{\psi}(q_1 r_1; q_2 r_2) \sim \sqrt{2E_{q_1} 2E_{q_2}} \langle q_1 r_1; q_2 r_2 | S - 1 | H \rangle$$

$$\begin{aligned} \tilde{\psi}(\vec{r}_1 s_1; \vec{r}_2 s_2) &= \int \frac{d^3 p_1 d^3 p_2}{(2\pi)^6 \sqrt{2E_{p_1} 2E_{p_2}}} \tilde{\psi} e^{i\vec{p}_1 \cdot \vec{r}_1 + i\vec{p}_2 \cdot \vec{r}_2} \\ &= \int \frac{d^3 p_1 d^3 p_2}{(2\pi)^6} \underbrace{\langle p_1 s_1; p_2 s_2 | S - 1 | H \rangle}_{(2\pi)^4 \delta^4(P_H - P_1 - P_2) i \frac{m_f}{v} \bar{u}(p_1 s_1) v(p_2 s_2)} e^{i\vec{p}_1 \cdot \vec{r}_1 + i\vec{p}_2 \cdot \vec{r}_2} \end{aligned}$$

Using the spherical system of coordinates

$$\int \frac{d^3 p_1 d^3 p_2}{(2\pi)^6} \delta^4(P_H - E_{p_1} - E_{p_2}) i \frac{m_f}{v} \bar{u}(p_1 s_1) v(p_2 s_2) \Big|_{\vec{p}_2 = -\vec{p}_1} e^{i\vec{p}_1 \cdot \vec{r}_1 + i\vec{p}_1 \cdot \vec{r}_2}$$

$$d^3 P_1 = \underbrace{P_1^2 dP_1}_{p_1 E_1 dE_1} d\Omega_{p_1}$$

$$E^2 = p^2 + m^2, \quad EdE = pdp$$

$$p_1^2 dp_1 = p_1 p_1 dp_1 = p_1 E_1 dE_1 = \sqrt{E_1^2 - m^2} E_1 dE_1$$

Using the property of delta function: $\int f(x) \delta(g(x)) = \sum_i \frac{f(x_i)}{|g'(x_i)|}$ and $g \equiv m_H - 2E_1 \equiv g(E_1)$, $g'(E_1) = -2$

$$\begin{aligned} & \frac{1}{(2\pi)^6} \int d\Omega dE_1 \sqrt{E_1^2 - m^2} \delta^4(m_H - 2E_1) i \frac{m_f}{v} \bar{u}(\vec{p}_1, s_1) v(-\vec{p}_1, s_2) \\ &= \frac{1}{(2\pi)^6} \frac{1}{2} \int d\Omega \sqrt{\left(\frac{m_H}{2}\right)^2 - m^2} \frac{m_H}{2} i \frac{m_f}{v} \bar{u}(\vec{p}_1, s_1) v(-\vec{p}_1, s_2) \Big|_{|\vec{p}_1| = \sqrt{\left(\frac{m_H}{2}\right)^2 - m^2}} \end{aligned}$$

We have reached this expression

$$\psi(\vec{r}_1 - \vec{r}_2; s_1, s_2) \sim \int d\theta \sin \theta d\phi \underbrace{f(\theta, \phi)}_{\bar{u}v} e^{i|\vec{p}| \underbrace{(\vec{r}_1 - \vec{r}_2) \cdot \hat{n}}_{\equiv \vec{r}}}$$

where $\hat{n} = \frac{\vec{p}}{|\vec{p}|}$

3 Fermion Antifermion product

From Dirac's equation for fermion

$$u(p) = \begin{pmatrix} \sqrt{E_p + m_f} \chi(\vec{s}_1) \\ \sqrt{E_p - m_f} (\vec{\sigma} \cdot \hat{n}_p) \chi(\vec{s}_2) \end{pmatrix}$$

and antifermion $v(p, s) = c\gamma^0 u^*(p, s)$

$$v(-p) = \begin{pmatrix} \sqrt{E_p - m_f} (\vec{\sigma} \cdot \hat{n}_{-p}) \eta(\vec{s}_2) \\ \sqrt{E_p + m_f} \eta(\vec{s}_2) \end{pmatrix}$$

taking in consideration $\hat{n}_{-p} = -\hat{n}_p$ and $\eta \hat{s}_2 = i\sigma_y \chi^*(\hat{s}_2)$

And for

$$\chi(\vec{s}_1) = \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{i\phi} \end{pmatrix}$$

we get

$$\eta(\vec{s}_2) = i\sigma_y \chi^*(\vec{s}_1) = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \cdot \begin{pmatrix} \cos \frac{\theta_2}{2} \\ \sin \frac{\theta_2}{2} e^{-i\phi_2} \end{pmatrix} = \begin{pmatrix} -i \sin \frac{\theta_2}{2} e^{-i\phi_2} \\ i \cos \frac{\theta_2}{2} \end{pmatrix}$$

Substitute in our

$$\bar{u} = u^\dagger \gamma^0 = (\sqrt{E_p + m_f} \chi^\dagger, -\sqrt{E_p - m_f} \chi^\dagger (\vec{\sigma} \cdot \hat{n}_p))$$

$$v = \begin{pmatrix} \sqrt{E_p - m_f} (\vec{\sigma} \cdot \hat{n}_{-p}) \eta(\vec{s}_2) \\ \sqrt{E_p + m_f} \eta(\vec{s}_2) \end{pmatrix} = \begin{pmatrix} -\sqrt{E_p - m_f} (\vec{\sigma} \cdot \hat{n}_p) \eta(\vec{s}_2) \\ \sqrt{E_p + m_f} \eta(\vec{s}_2) \end{pmatrix}$$

$$\bar{u}v = -2\sqrt{(E_p + m_f)(E_p - m_f)} \chi^\dagger (\vec{\sigma} \cdot \hat{n}_p) \eta$$

$$\chi^\dagger = (\cos \frac{\theta}{2}, \sin \frac{\theta}{2} e^{-i\phi}) \eta = \begin{pmatrix} -i \sin \frac{\theta}{2} e^{-i\phi} \\ i \cos \frac{\theta}{2} \end{pmatrix}$$

Expression for

$$\vec{\sigma} \cdot \hat{n}_p = \sigma_x n_x + \sigma_y n_y + \sigma_z n_z = \begin{pmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{pmatrix}$$

$$\hat{n}_p = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$$

$$\vec{s}_i = (\sin \theta_i \cos \phi_i, \sin \theta_i \sin \phi_i, \cos \theta_i)$$

$$\vec{\sigma} \cdot \hat{n}_p = \begin{pmatrix} \cos \theta & \sin \theta \cos \phi - i \sin \theta \sin \phi \\ \sin \theta \cos \phi + i \sin \theta \sin \phi & -\cos \theta \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix}$$

$$(\vec{\sigma} \cdot \hat{n}_p) \eta = \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix} \begin{pmatrix} -i \sin \frac{\theta}{2} e^{-i\phi} \\ i \cos \frac{\theta}{2} \end{pmatrix} = \begin{pmatrix} -i \cos \theta \sin \frac{\theta}{2} e^{-i\phi} + i \sin \theta \cos \frac{\theta}{2} e^{-i\phi} \\ -i \sin \theta \sin \frac{\theta}{2} + i \cos \theta \cos \frac{\theta}{2} \end{pmatrix} = \begin{pmatrix} i \sin \frac{\theta}{2} \\ i \cos \frac{3\theta}{2} \end{pmatrix}$$

(Could be mistake here - **CHECK**)

$$\begin{aligned} \chi^\dagger(\vec{s}_1) \hat{n} \vec{\sigma} \eta(\vec{s}_2) &= \left[\cos \frac{\theta_1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \sin \frac{\theta_1}{2} e^{-i\phi_1} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right] \\ &\times \left[\cos \frac{\theta_2}{2} \begin{pmatrix} \cos \theta_p \\ \sin \theta_p \end{pmatrix} + \sin \frac{\theta_2}{2} e^{i\phi_2} \begin{pmatrix} \sin \theta_p e^{-i\phi} \\ -\cos \theta_p \end{pmatrix} \right]^T \\ &= \cos \frac{\theta_1}{2} \cos \frac{\theta_2}{2} \cos \theta_p + \cos \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i\phi_2} \sin \theta_p e^{-i\phi_p} \\ &+ \sin \frac{\theta_1}{2} \cos \frac{\theta_2}{2} \sin \theta_p e^{-i\phi_p} e^{-i\phi_1} - \sin \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i(\phi_2 - \phi_1)} \cos \theta_p \end{aligned}$$

4 Calculating integral in spherical coordinates

$$\int d\hat{n} \chi^\dagger(\vec{s}_1) \hat{n} \vec{\sigma} \eta(\vec{s}_2) = \int_0^{2\pi} d\phi_p \int_0^\pi d\theta_p \sin \theta_p e^{i\vec{p} \cdot \vec{r}} \times$$

$$\left[\cos \frac{\theta_1}{2} \cos \frac{\theta_2}{2} \cos \theta_p + \cos \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i\phi_2} \sin \theta_p e^{-i\phi_p} + \sin \frac{\theta_1}{2} \cos \frac{\theta_2}{2} \sin \theta_p e^{i\phi_p} e^{-i\phi_1} - \sin \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i(\phi_2 - \phi_1)} \cos \theta_p \right]$$

Some expressions for spherical harmonics

$$Y_0^0 = \frac{1}{2} \sqrt{\frac{1}{\pi}} \quad Y_1^1 = -\frac{1}{2} \sqrt{\frac{3}{2\pi}} \sin \theta e^{i\phi} \quad (1)$$

$$Y_1^0 = \frac{1}{2} \sqrt{\frac{3}{\pi}} \cos \theta \quad Y_1^{-1} = \frac{1}{2} \sqrt{\frac{3}{2\pi}} \sin \theta e^{-i\phi} \quad (2)$$

$$Y_\ell^{m*}(\theta, \varphi) = (-1)^{-m} Y_\ell^{-m}(\theta, \varphi) \quad (3)$$

Using the expression for $e^{i\vec{p} \cdot \vec{r}}$ and substituting a trigonometric function for the corresponding spherical harmonic

$$\int_0^{2\pi} d\phi_p \int_0^\pi d\theta_p \sin \theta_p e^{i\vec{p} \cdot \vec{r}} \sum_{l=0}^{\infty} \sum_{m=-l}^l 4\pi i^l j_l(kr) Y_l^m(\theta_p, \varphi_p) Y_l^{m*}(\theta_r, \varphi_r) \times$$

$$\left[\cos \frac{\theta_1}{2} \cos \frac{\theta_2}{2} \cdot 2\sqrt{\frac{\pi}{3}} Y_0^1(\theta_p, \phi_p) \right.$$

$$+ \cos \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i\phi_2} Y_1^{-1}(\theta_p, \phi_p)$$

$$+ (-2) \sqrt{\frac{2\pi}{3}} \sin \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i\phi_1} Y_1^1(\theta_p, \phi_p)$$

$$\left. + (-2) \sqrt{\frac{\pi}{3}} \sin \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i(\phi_2 - \phi_1)} Y_1^0(\theta_p, \phi_p) \right]$$

Applying orthogonality $\int_{\theta=0}^\pi \int_{\varphi=0}^{2\pi} Y_\ell^m Y_{\ell'}^{m'*} d\Omega = \delta_{\ell\ell'} \delta_{mm'}$,

(**MISTAKE** I forgot that orthogonality uses $Y_1^{-1*}(\theta_p, \varphi_p)$ instead of $Y_1^{-1}(\theta_p, \varphi_p)$). Must

add corresponding $(-1)^m$ to each term)

$$\begin{aligned}
4\pi j_1(kr) \int_0^{2\pi} d\phi_p \int_0^{2\pi} d\theta_p \sin \theta_p & \left[2\sqrt{\frac{\pi}{3}} \cos \frac{\theta_1}{2} \cos \frac{\theta_2}{2} \right. \\
& \times \underbrace{Y_1^0(\theta_p, \varphi_p) Y_1^0(\theta_p, \varphi_p)}_{=1} Y_1^{0*}(\theta_r, \varphi_r) \\
& + 2\sqrt{\frac{\pi}{3}} \cos \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i\phi_2} \\
& \times \underbrace{Y_1^{-1}(\theta_p, \varphi_p) Y_1^{-1}(\theta_p, \varphi_p)}_{=1} Y_1^{-1*}(\theta_r, \varphi_r) \\
& + (-2)\sqrt{\frac{2\pi}{3}} \sin \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i\phi_1} \\
& \times \underbrace{Y_1^1(\theta_p, \varphi_p) Y_1^1(\theta_p, \varphi_p)}_{=1} Y_1^{1*}(\theta_r, \varphi_r) \\
& + (-2)\sqrt{\frac{\pi}{3}} \sin \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i(\phi_2 - \phi_1)} \\
& \times \underbrace{Y_1^0(\theta_p, \varphi_p) Y_1^0(\theta_p, \varphi_p)}_{=1} Y_1^{0*}(\theta_r, \varphi_r) \left. \right]
\end{aligned}$$

$$\begin{aligned}
4\pi j_1(kr) \int_0^{2\pi} d\phi_p \int_0^{2\pi} d\theta_p \sin \theta_p & \left[2\sqrt{\frac{\pi}{3}} \cos \frac{\theta_1}{2} \cos \frac{\theta_2}{2} \right. \\
& \times \underbrace{Y_1^0(\theta_p, \varphi_p) Y_1^{0*}(\theta_p, \varphi_p)}_{=1} Y_1^{0*}(\theta_r, \varphi_r) \\
& - 2\sqrt{\frac{\pi}{3}} \cos \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i\phi_2} \\
& \times \underbrace{Y_1^{-1}(\theta_p, \varphi_p) Y_1^{-1*}(\theta_p, \varphi_p)}_{=1} Y_1^{-1*}(\theta_r, \varphi_r) \\
& - (-2)\sqrt{\frac{2\pi}{3}} \sin \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i\phi_1} \\
& \times \underbrace{Y_1^1(\theta_p, \varphi_p) Y_1^{1*}(\theta_p, \varphi_p)}_{=1} Y_1^{1*}(\theta_r, \varphi_r) \\
& + (-2)\sqrt{\frac{\pi}{3}} \sin \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i(\phi_2 - \phi_1)} \\
& \times \underbrace{Y_1^0(\theta_p, \varphi_p) Y_1^{0*}(\theta_p, \varphi_p)}_{=1} Y_1^{0*}(\theta_r, \varphi_r) \left. \right]
\end{aligned}$$

5 Using the dot products

$$\begin{aligned} & \left[\left(-2\sqrt{\frac{\pi}{3}} \cos \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i\phi_2} \right) Y_1^{*-1}(\theta_r, \phi_r) \right. \\ & + \left(2\sqrt{\frac{\pi}{3}} \cos \frac{\theta_1}{2} \cos \frac{\theta_2}{2} - 2\sqrt{\frac{\pi}{3}} \sin \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i(\phi_2 - \phi_1)} \right) Y_1^{*0}(\theta_r, \phi_r) \\ & \left. + \left(2\sqrt{\frac{2\pi}{3}} \sin \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i\phi_1} \right) Y_1^{*1}(\theta_r, \phi_r) \right] \end{aligned}$$

6 Conclusion